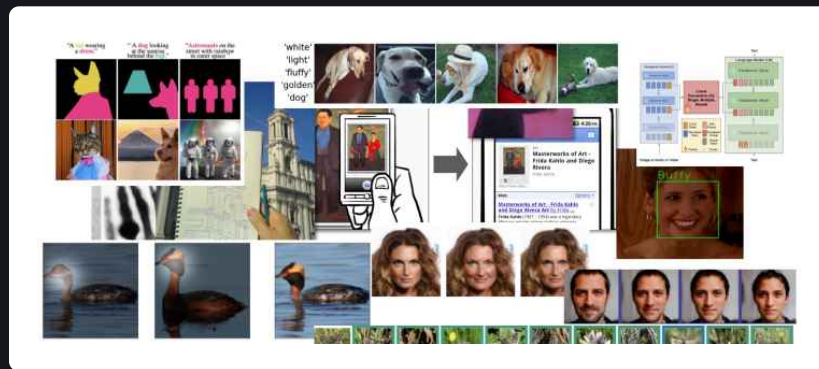


Introduction to computer vision and machine learning

Sorbonne University

clementrambour.github.io/teaching-rdfia



Course outline

01 **Intro to computer vision and machine learning**

Today

02 Intro to neural networks + ML theory

03 Neural nets for image classification

04 Large ConvNets

05 Vision transformers

06 Segmentation, transfer learning, domain adaptation

07 Vision-language models

08 Explaining VLMs

09 Self-supervised learning in vision

10 Generative models with GANs

11 **Control**

Jan 07, 2027

12 Diffusion models

13 Bayesian deep learning

14 Uncertainty, robustness

UE Information on

`clementrambour.github.io/teaching-rdfia.html`
`https://rdfia.github.io/`

Mattermost: M2 MIND `http://tiny.cc/M2MIND26`

RDFIA Channel: `https://channel.isir.upmc.fr/m1-mind-25-27/channels/rdfia`

Evaluation

30%

Control

Written exam, Jan 07, 2027

30%

Practicals • 3 reports

Each report: PDF + code

30%

Continual assessment

Short written exam at the end
of dedicated lab sessions

SECOND CHANCE EXAM

Replaces the previous grade (control + reports + continual assessment)

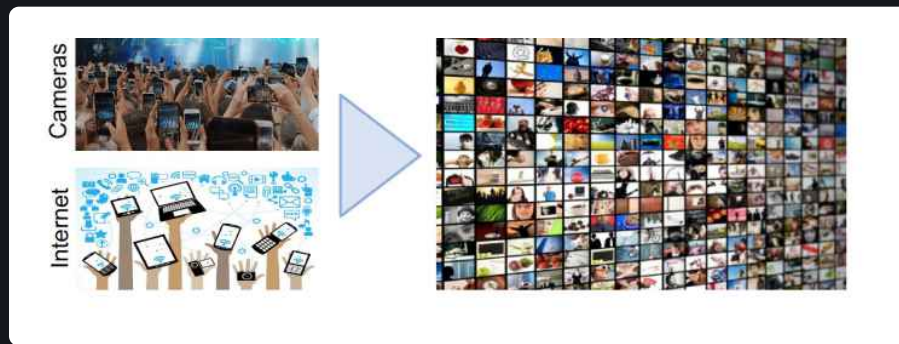
INTRODUCTION

Computer vision

(Processing, analyzing and)
understanding visual data

Where are we now?

Source (many slides): Cornell CV course



Optical character recognition (OCR)

- Digit recognition, AT&T Labs
- License plate readers
- Automatic check processing
- If you have a scanner, it probably came with OCR software

- If you have a scanner, it probably came with OCR software



Digit recognition, AT&T labs
<http://www.research.att.com/~yannf/>



License plate readers
http://en.wikipedia.org/wiki/Automatic_number_plate_recognition



DEPLOYED

Face detection

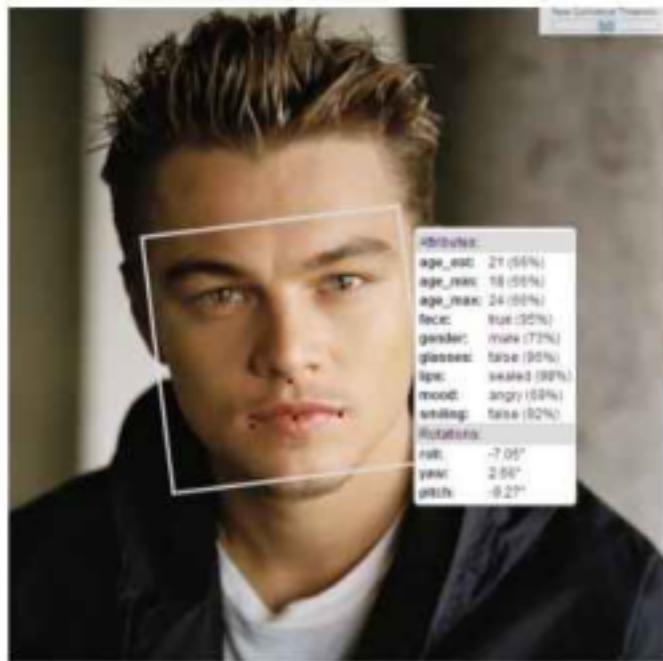
- Cameras now detect faces —
Canon, Sony, Fuji, ...



- Cameras now detect faces

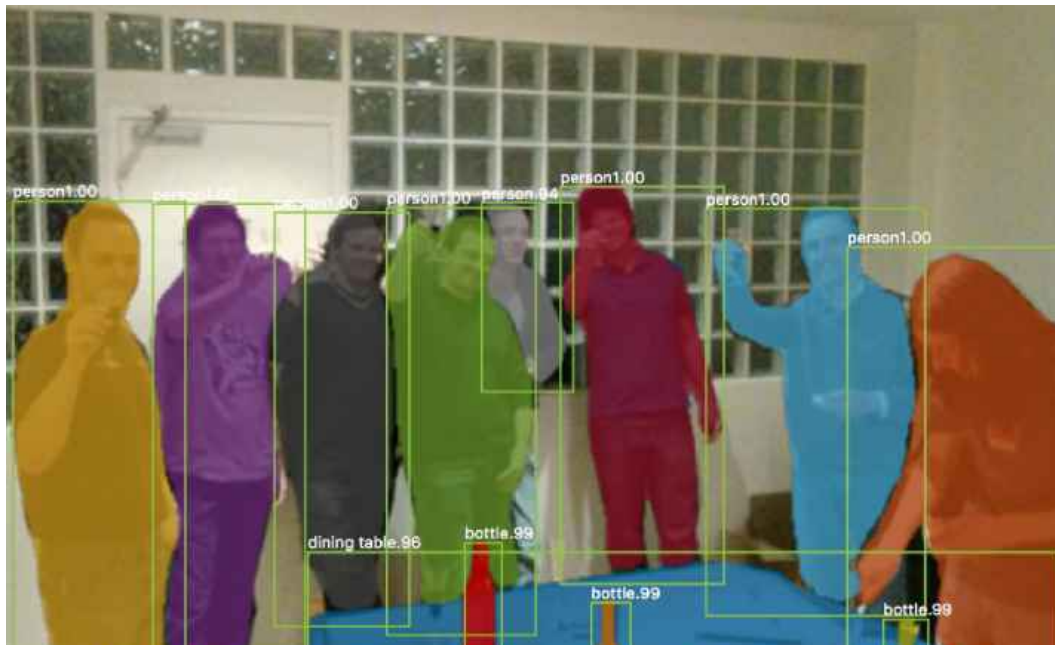
DEPLOYED • SIGNIFICANT PROGRESS

Face recognition



SIGNIFICANT PROGRESS

Recognizing objects



Mask R-CNN. K. He, G. Gkioxari, P. Dollár, R. Girshick. ICCV 2017

EXAMPLE

Recognition-based product search



EXAMPLE

Recognition-based product search



EXAMPLE

Recognition-based product search



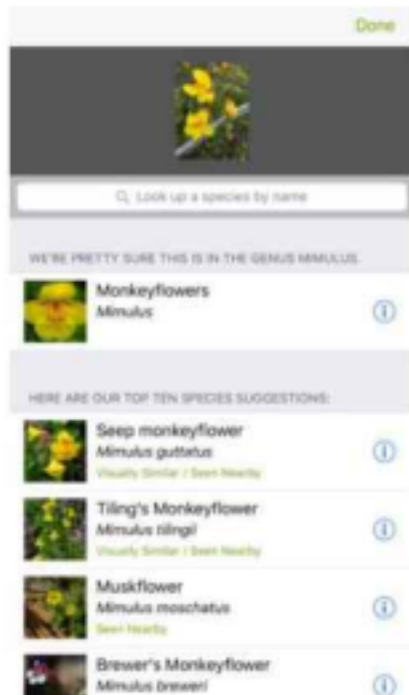
SIGNIFICANT PROGRESS

Species recognition

iNaturalist dataset

CHALLENGES

- Fine-grained recognition
- Detecting rare concepts



iNaturalist dataset

Challenges:

- fine-grained recognition
- Detecting rare concepts

CHALLENGES

Fully autonomous driving



Medical imaging, health

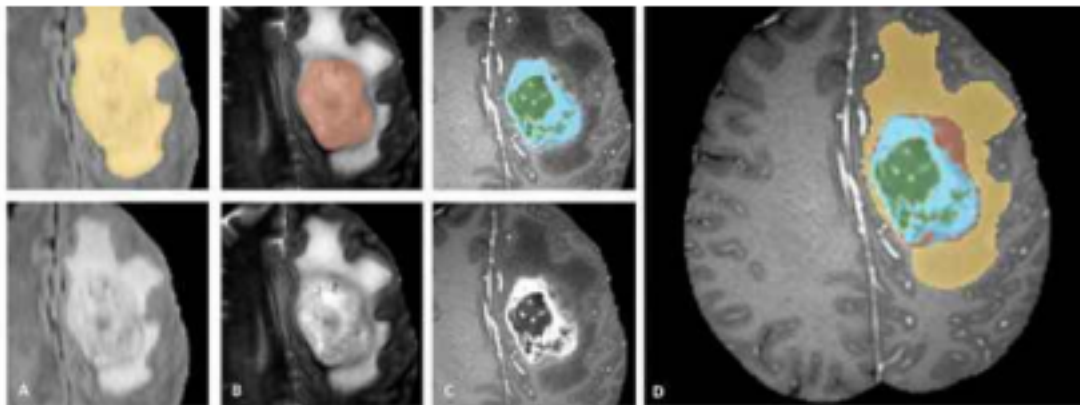


Fig.1: Glioma sub-regions. Shown are image patches with the tumor sub-regions that are annotated in the different modalities (top left) and the final labels for the whole dataset (right). The image patches show from left to right: the whole tumor (yellow) visible in T2-FLAIR (Fig.A), the tumor core (red) visible in T2 (Fig.B), the enhancing tumor structures (light blue) visible in T1Gd, surrounding the cystic/necrotic components of the core (green) (Fig. C). The segmentations are combined to generate the final labels of the tumor sub-regions (Fig.D): edema (yellow), non-enhancing solid core (red), necrotic/cystic core (green), enhancing core (blue). (Figure taken from the [BraTS IEEE TMI paper](#).)

CHALLENGES

Medical imaging, health

Building a system to detect Covid in chest X-rays

What should a metric measure?

$$\text{Accuracy} = P(\text{pred_label} = \text{true_label})$$

95%

Accuracy of the candidate system

Is this good?

Did it actually help / work?

Why?

CHALLENGES

Typical issues that plague deployment

Domain shift

Images seen during deployment are very different

Concept drift

Meaning of classes etc. changes

Open world

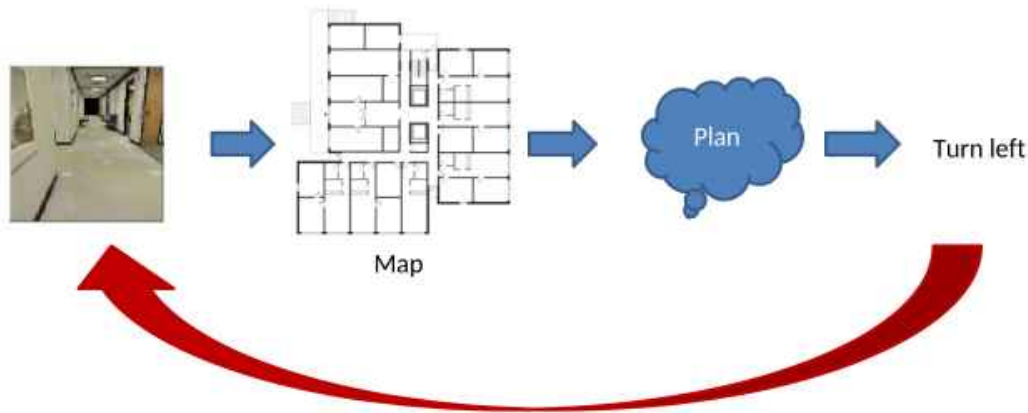
Unforeseen circumstances, e.g. new classes

- Unforeseen circumstances, e.g., new classes; open world



CHALLENGES

Integrating vision and action, robotics



Cognitive Mapping and Planning for Visual Navigation

S. Gupta, J. Davidson, S. Levine, R. Sukthankar, J. Malik.

Cognitive Mapping and Planning for Visual Navigation. S. Gupta, J. Davidson, S. Levine, R. Sukthankar, J. Malik.
CVPR 2017

CHALLENGES

Understanding complex situations, reasoning



VQA task: why is this funny?

VQA task: why is this funny?



Andrej Karpathy

VQA TASK · CHATGPT-40 ANSWER

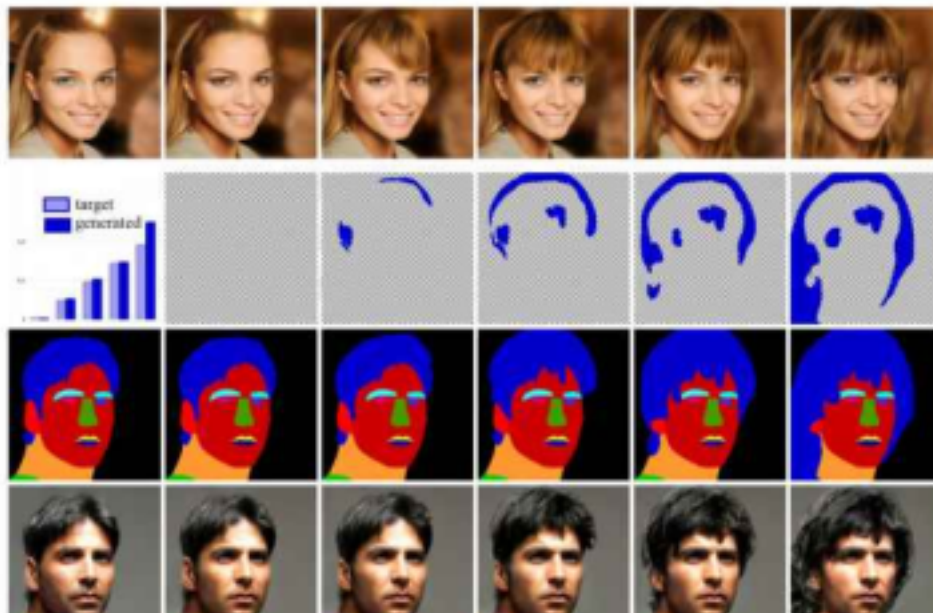
Why is this funny?



“The humor in this image stems from the playful interaction between former President Barack Obama and a much taller man who appears to be getting his height measured. The scene is lighthearted and unexpected, showing a humorous side of a usually serious figure like the President. The contrast in their heights, with Obama leaning back and smiling while the taller man stands on a scale, adds to the comedic effect. The expressions and body language of the other men in the background, who are also smiling and laughing, contribute to the overall jovial atmosphere.”

CHALLENGES

Generative models for image editing and manipulation (GANs)



CHALLENGES

Image generation from text (diffusion models, 2023)

Sprouts in the shape of text
'Imagen' coming out of a fairytale
book



A photo of a Shiba Inu dog with a
backpack riding a bike. It is
wearing sunglasses and a beach
hat.



A high contrast portrait of a very
happy fuzzy panda dressed as a
chef in a high end kitchen making
dough. There is a painting of
flowers on the wall behind him.



COURSE OUTLINE • CHAPTER 1

Computer vision and machine learning basics

A Visual (local) feature detection

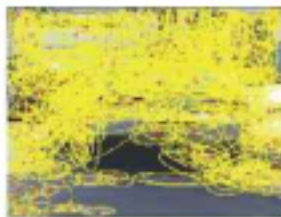
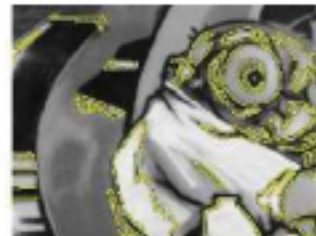
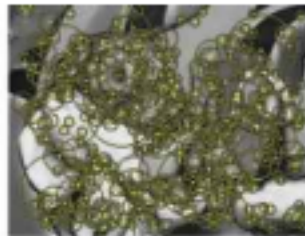
Now

B Visual (local) feature description

C Bag-of-Words image representation

Local feature detection and description

- Points / regions of interest detection
- One example: corner detection (Harris corner detector)



**Sparse, at
interest points**



Dense, uniformly



Randomly

One example: Corner detection (Harris corner detector)

Corner detection

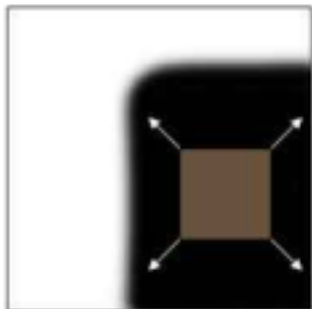
CORNER POINT

A singular point: highly informative, rare, ...

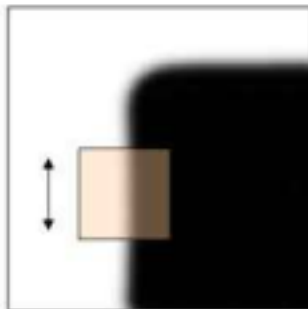
BASIC IDEA FOR THE ALGORITHM

For each pixel (x, y) of image I , translate a centered window. If (x, y) is a corner, it should cause **large differences in patch appearance** — whatever the translation.

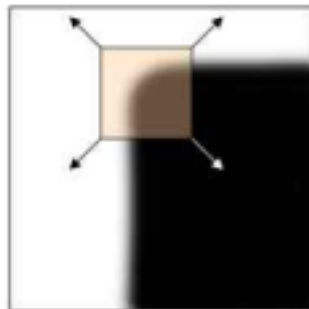
Corner detection: basic idea



"flat" region:
no change in
all directions



"edge":
no change along
the edge
direction



"corner":
significant
change in all
directions

Corner detection op == For all pix, shift a window in *any direction*, keep the ones that give a *large change* in intensity

Corner detection

Consider a small window W and a shifting vector (u, v)

- How do the pixels (x, y) in W change?
- Compare each pixel before and after by summing up the squared differences (SSD)
- If $E(u, v)$ is high, the center of W is a corner candidate

Repeat for all shifting directions (u, v) . The center of W is a corner if $E(u, v)$ is high for all (u, v) .

$$E(u, v) = \sum_{(x, y) \in W} [I(x + u, y + v) - I(x, y)]^2$$

Very computationally expensive → simplified to get the Harris corner detector

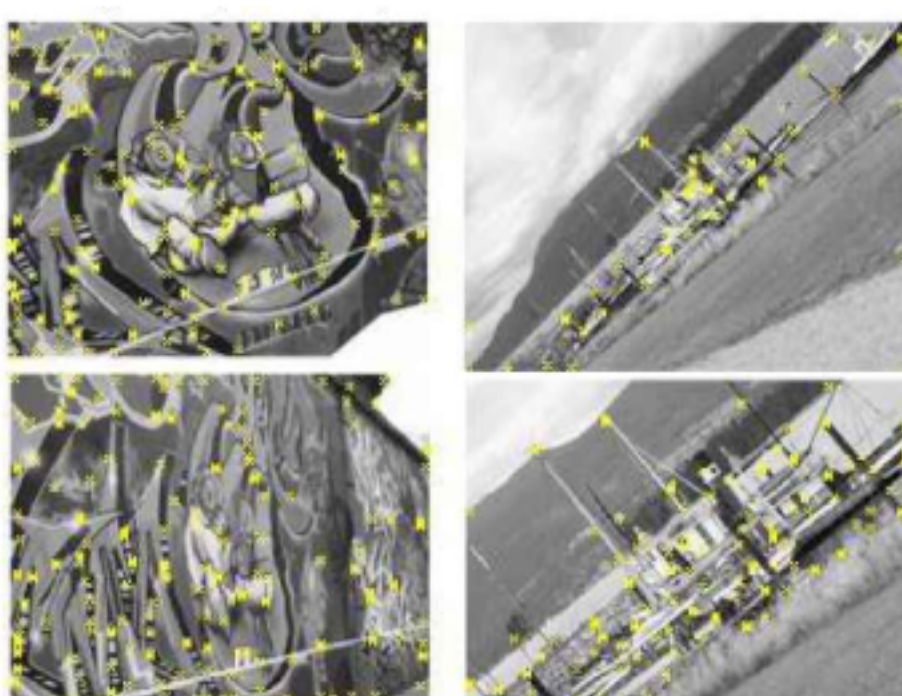
Harris detector example



Harris features (in red)

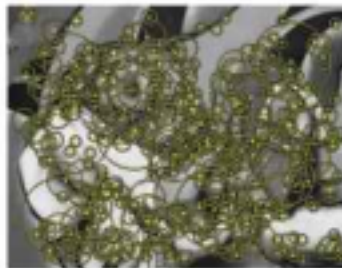


Looking for repeatability



Many other point / region of interest detectors

Many other Points/Regions of Interest detectors



Sparse, at



Dense, uniformly



Randomly

COURSE OUTLINE • CHAPTER 1

Computer vision and machine learning basics

A Visual (local) feature detection

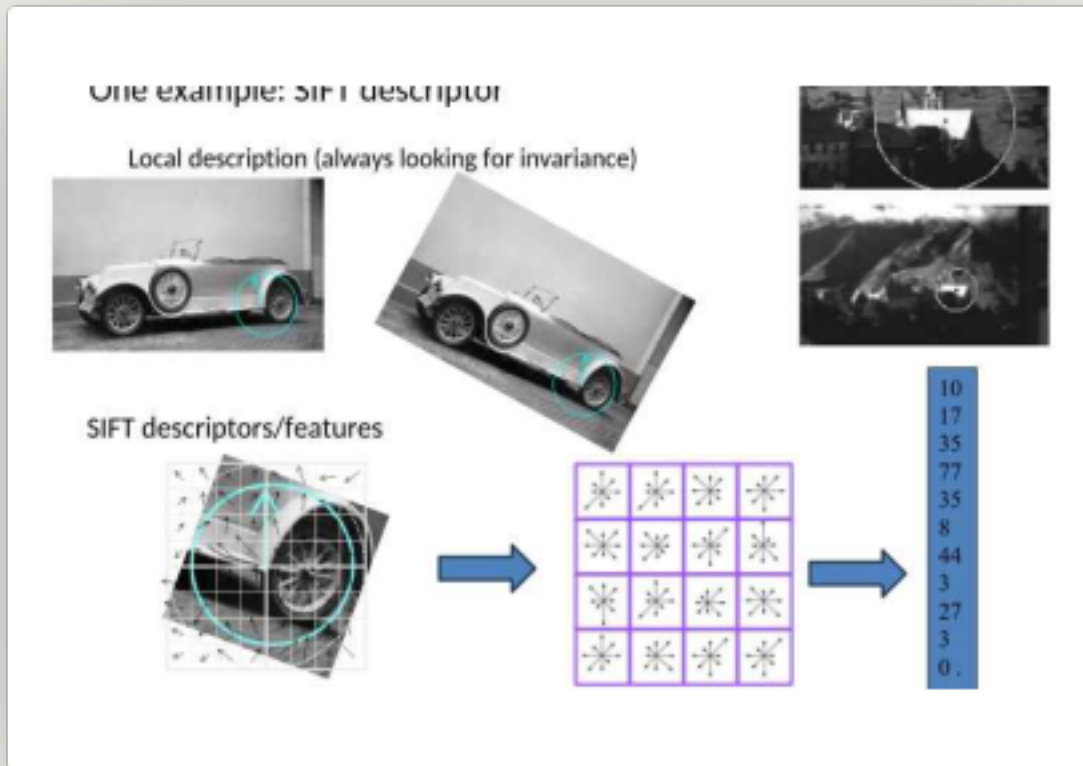
B Visual (local) feature description

Now

c Bag-of-Words image representation

Local feature description

- Local description: always looking for invariance
- Many point / region of interest descriptors
- One example: the SIFT descriptor



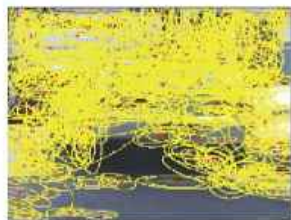
Feature descriptors

EXPECTED PROPERTIES

- Similar patches → close descriptors
- Invariance (robustness) to geometric transformations: rotation, scale, viewpoint, luminance, semantics?



BoF: a first image representation



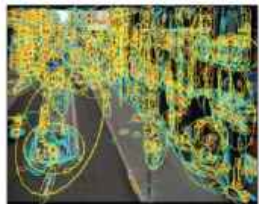
Sparse, at
interest points



Dense, uniformly



Randomly



Multiple interest
operators

Feature extraction



A bag of features
BoF

Bag of Features (BoF) model

Image



(features)



Image representation

BoF

BAG OF FEATURES

- Local signatures: not a scalable representation
- Not a semantic representation



BoW

BAG OF (VISUAL) WORDS

The missing bit: **the visual word**

COURSE OUTLINE • CHAPTER 1

Computer vision and machine learning basics

A Visual (local) feature detection

B Visual (local) feature description

C Bag-of-Words image representation Now

COURSE OUTLINE · C

Bag-of-Words image representation

c1 Introduction to Bag of Words

Now

c2 Visual dictionary

c3 Image signature

c4 Whole recognition pipeline

BoW model: textual representation and color indexing

Comparing two documents
using visual / color / word
occurrences

representation and color indexing



Comparing 2 docs using visual/color/word occurrences

Slide credit: L. Fei-Fei

Bag of Visual Words (BoW)

BoW: a histogram over a visual dictionary of features

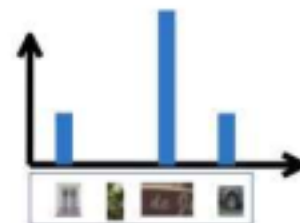
QUESTIONS

- 1 Which dictionary?
- 2 How to project the BoF onto the dictionary?
- 3 How to compute the histogram?

(features)



BoW : histogram on visual dictionary



COURSE OUTLINE · C

Bag-of-Words image representation

c1 Introduction to Bag of Words

c2 Visual dictionary

Now

c3 Image signature

c4 Whole recognition pipeline

Visual space clustering

- 1 Extraction of local features (patterns / visual words) in images — training set for classification, image dataset for retrieval
- 2 Clustering of the feature space

1. Extraction of local features (pattern/visual words) in images
 - Training dataset in classification
 - Image dataset in retrieval
2. Clustering of feature space



Training set but no labels => UNSUPERVISED Learning

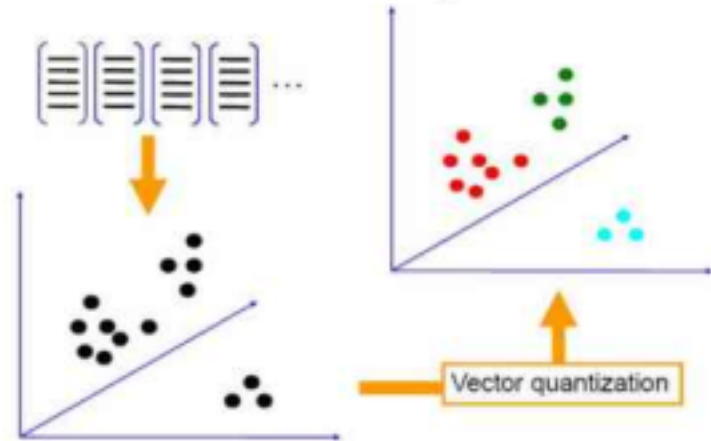
Training set but no labels →
unsupervised learning

Visual space clustering

Many algorithms for clustering:

- K-means
- Vector quantization
- Gaussian mixture models
- ...

- Many algorithms for clustering :
 - K-Means
 - Vectorial Quantization
 - Gaussian Mixture Models
 - ...



Clustering with K clusters

INPUT

Set of n points $\{x_i\}_n$ in $\mathbb{R}^d$

GOAL

Find a set of K ($K \ll n$) points $w = \{w_k\}_K$ that approximates the n input points, i.e. minimizes the error criterion $C(w)$

$$C(w) = \sum_{i=1}^n \min_k \|x_i - w_k\|^2$$

At fixed K , complexity is $O(n^{(Kd+1)} \log n)$

There are many strategies to approximate the global optimization problem.

Clustering with the K-means algorithm

$$C(w) = \sum_{i=1}^n \min_k \|x_i - w_k\|^2$$

Init K centers $\{c_k\}$ by sampling K points w_k in $\mathbb{R}^d$

1. (Re)assign each point x_i to the cluster s_i with the center w_{s_i} so that $\text{dist}(x_i, w_{s_i})$ is less than dist from x_i to any other current cluster center.

$$\min_k \|x_i - w_k\|^2$$
2. Move all w_k inside each cluster as the new barycenter from all the points assigned to the cluster k (equivalent to minimizing the corresponding mean square error).

$$\sum_{i=1}^n \|x_i - w_{s_i}\|^2$$
3. Go to step 1 if some points changed clusters during the last iteration.

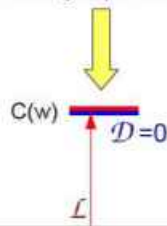
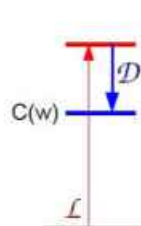
Output: The set of the final K cluster centers $\{c_k = w_k\}$

K-means: why is it successful?

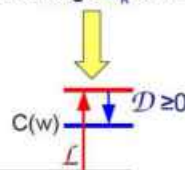
Consider an arbitrary cluster assignment s_i .

$$C(w) = \sum_{i=1}^n \min_k \|x_i - w_k\|^2 = \underbrace{\sum_{i=1}^n \|x_i - w_{s_i}\|^2}_{\mathcal{L}(s,w)} - \underbrace{\sum_{i=1}^n \|x_i - w_{s_i}\|^2 - \min_k \|x_i - w_k\|^2}_{\mathcal{D}(s,w) \geq 0}$$

1. Change s_i to minimize $\mathcal{D}$ leaving $C(w)$ unchanged.



2. Change w_k to minimize $\mathcal{L}$. Meanwhile $\mathcal{D}$ can only increase.



© L. Bottou

Clustering with K-means: pros and cons

PROS

Simplicity

Convergence (local minimum)

CONS

Memory-intensive

Depends on K

Sensitive to initialization

Sensitive to artifacts

Limited to spherical clusters

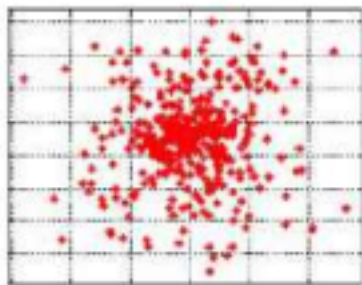
Clusters concentrate in high-density areas

Alternatives: radius-based methods

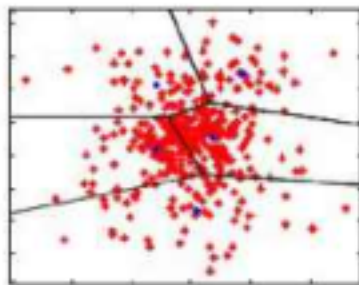
K-means is deeply used in practice

Clustering: uniform, K-means, radius-based

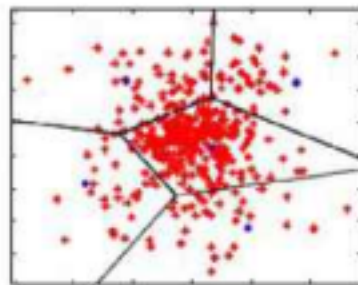
- Uniform / K-means / radius-based :



(a) Histogram



(b) K-means



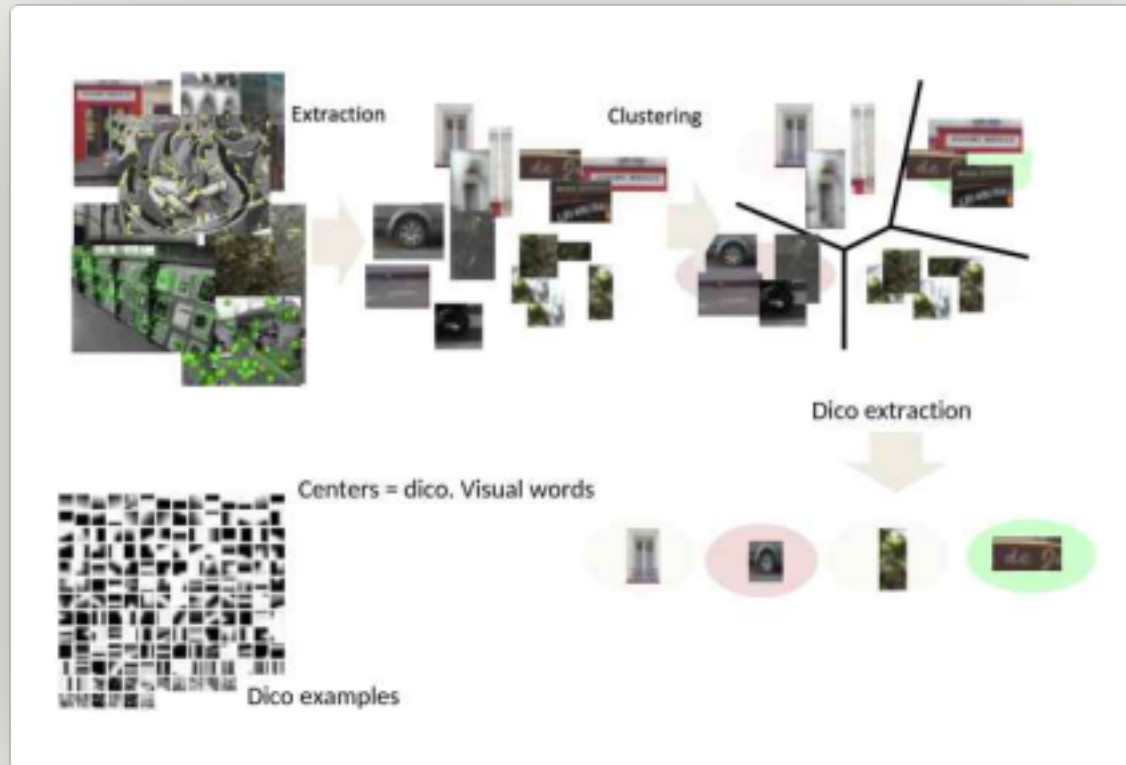
(c) Radius-based

- *Radius-based clustering assigns all features within a fixed radius of similarity r to one cluster.*

Radius-based clustering assigns all features within a fixed radius of similarity r to one cluster.

Dictionary = K visual words

- Cluster centers = dictionary of visual words
- Dictionary extraction and examples



COURSE OUTLINE · C

Bag-of-Words image representation

c1 Introduction to Bag of Words

c2 Visual dictionary

c3 **Image signature**

Now

c4 Whole recognition pipeline

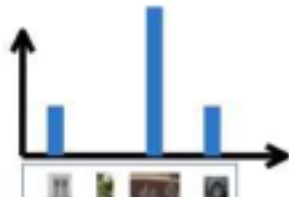
Bag-of-Words (BoW) image signature

FOR EACH IMAGE

- For each local feature: find the closest visual word
- Increase the corresponding bin in the histogram of the visual dictionary

IMAGE SIGNATURE (GLOBAL INDEX)

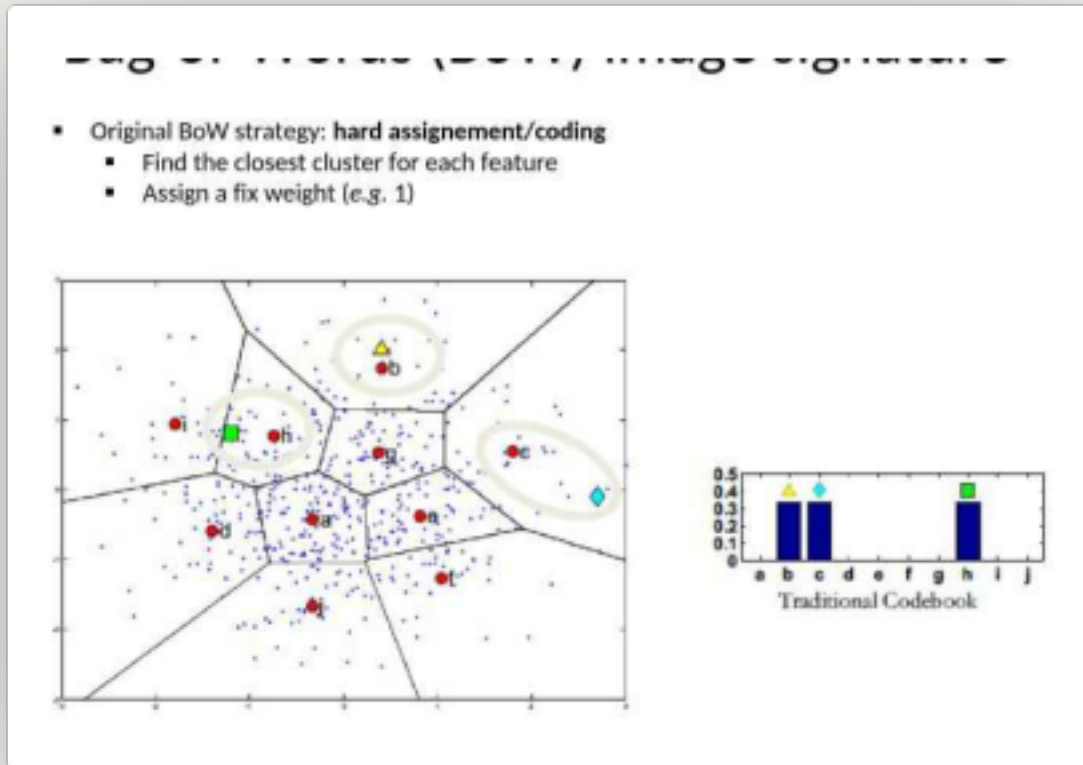
- Vector (histogram) of M bins, $M = K =$ dictionary size
- Each term represents a likelihood to get this visual word



Hard assignment coding

Original BoW strategy

- 1 Find the closest cluster for each feature
- 2 Assign a fixed weight (e.g. 1)



Sum pooling

Initial BoW strategy: just counting occurrences of words in the document

Classical BoW = **hard coding** + **sum pooling**

Sum pooling : initial BoW strategy (just counting occurrences of words in the document)

Classical BoW = **hard coding** + **sum pooling**

1. Find the closest cluster for each feature
2. Assign a fix weight (e.g. 1) to this cluster

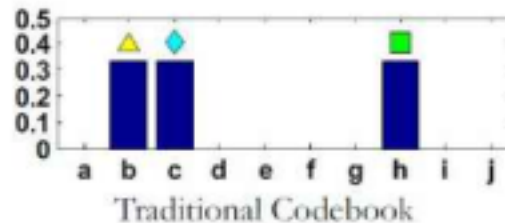
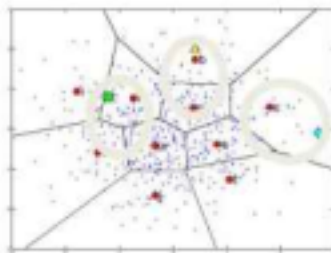
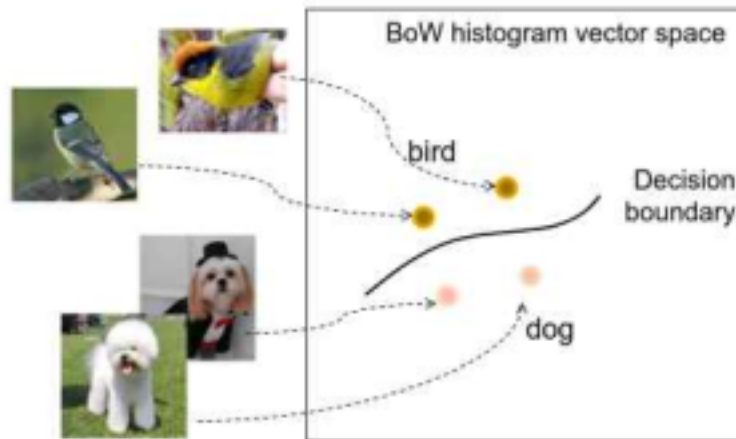


Image classification based on BoW

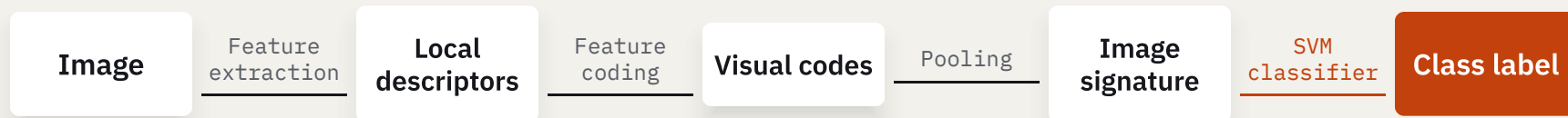
Images live as points in the BoW histogram vector space

Learn a classification model to determine the decision boundary



Learn a classification model to determine the decision boundary

A classification model to determine the decision boundary



SVM

- Images / patterns $\mathbf{x} \in \mathbf{X}$
- Φ : function transforming the patterns into feature vectors $\Phi(\mathbf{x})$ (like BoW, or on top of it)
- $\langle \cdot, \cdot \rangle$: dot product in the feature space endowed by $\Phi(\cdot)$
- SVM: binary classifier, classes $y = \pm 1$

Taking the sign of a linear discriminant function:

$$f(\mathbf{x}) = \langle \mathbf{w}, \Phi(\mathbf{x}) \rangle + b$$

SVM: how to find / estimate f ?

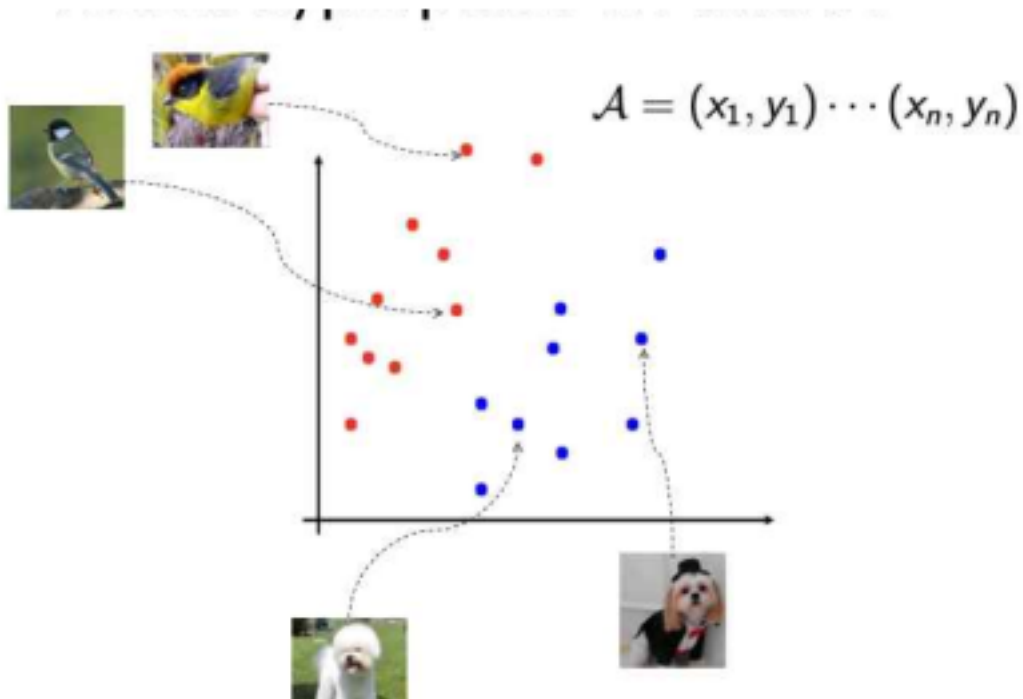
- Feature function Φ usually hand-chosen for each problem
- Several Φ for image processing, like BoW
- w and b : parameters to be determined

$$f(x) = \langle w, \Phi(x) \rangle + b$$

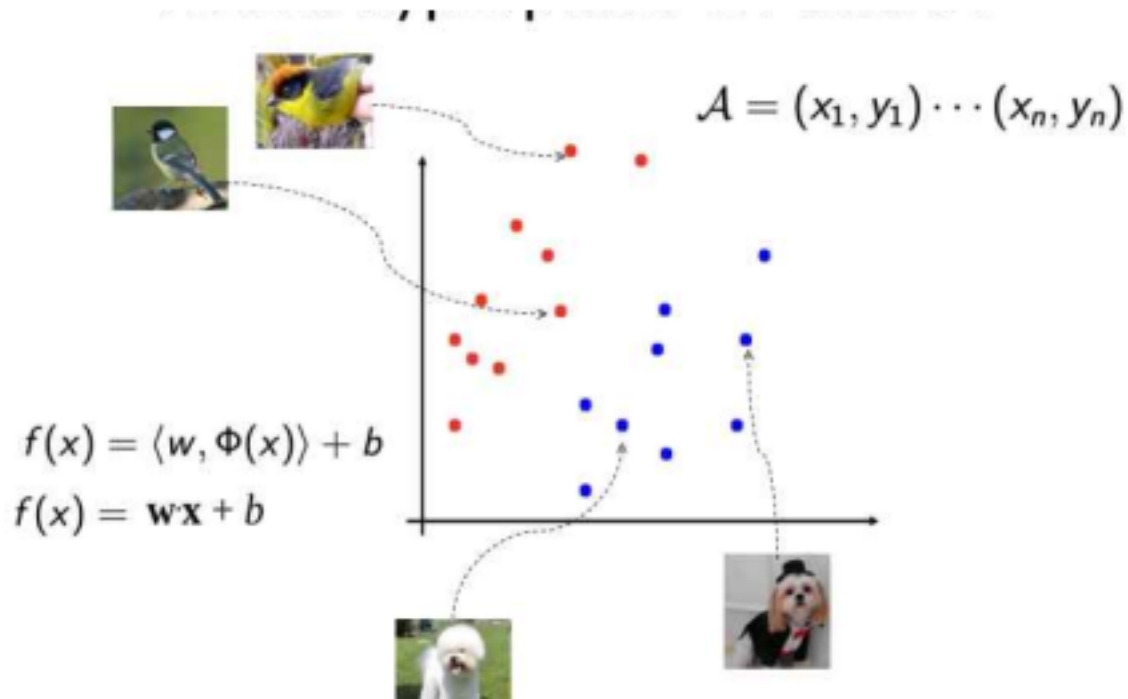
Learning algorithm on a set of training examples:

$$\mathcal{A} = (x_1, y_1) \cdots (x_n, y_n)$$

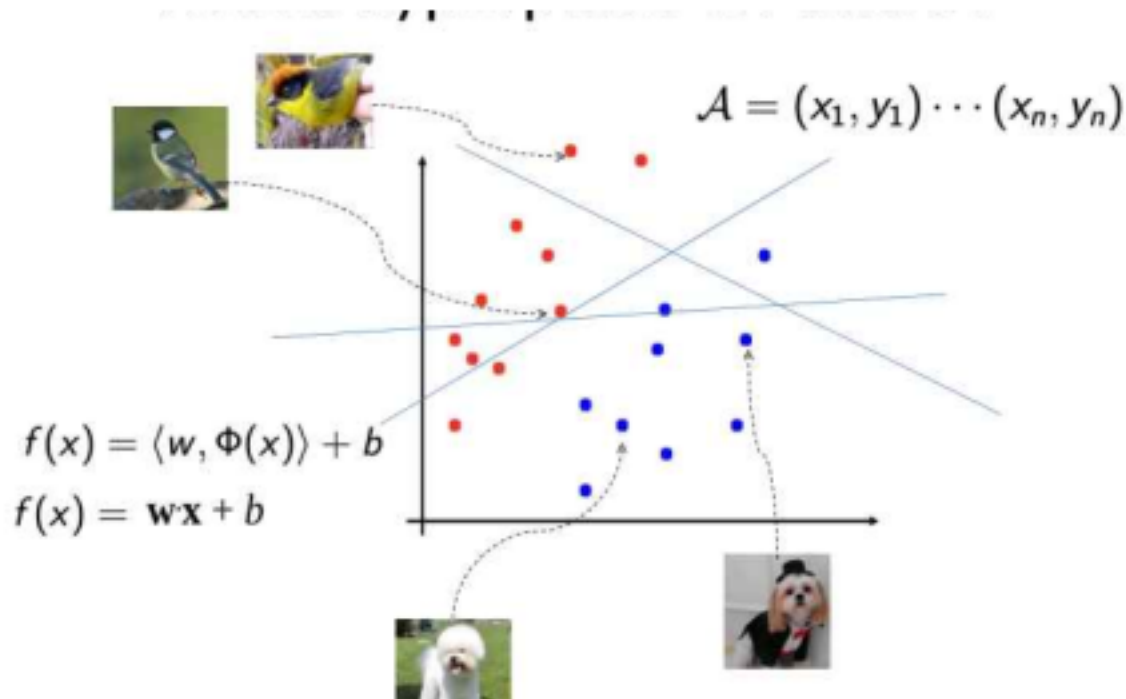
Which hyperplane w and b ?



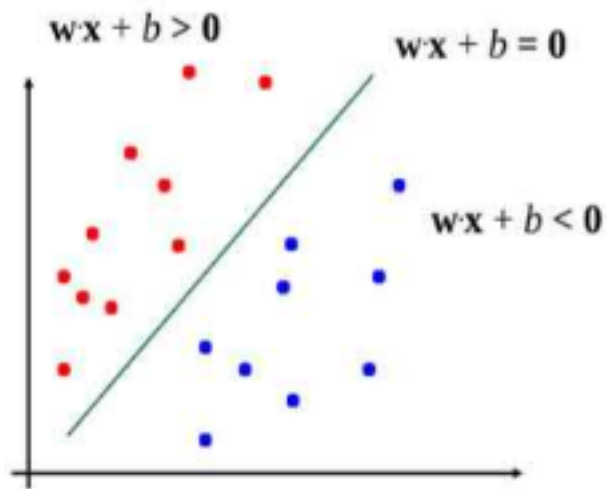
Which hyperplane w and b ?



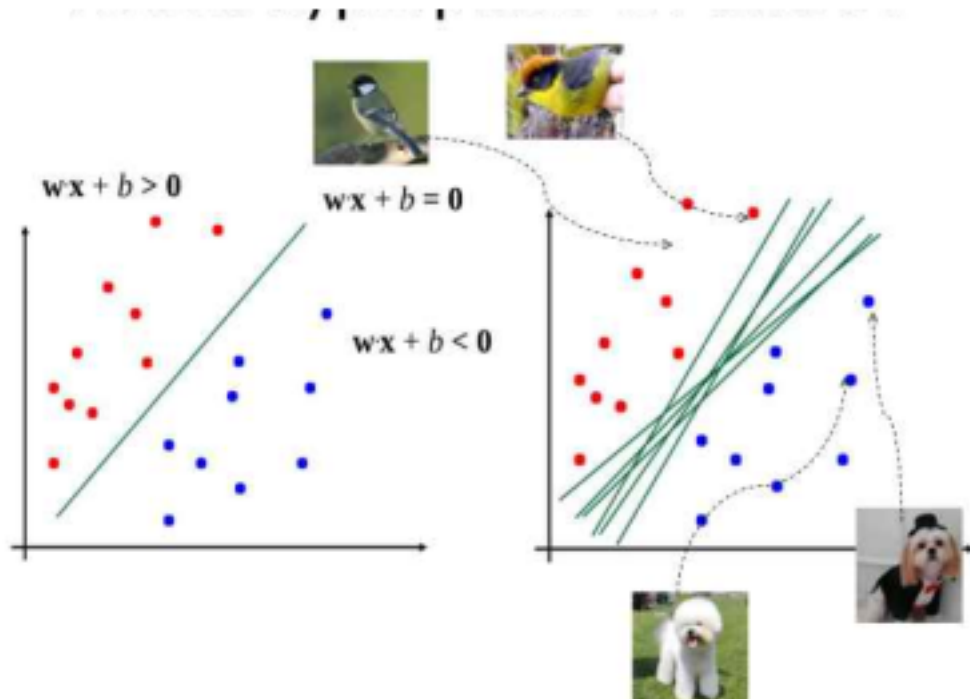
Which hyperplane w and b ?



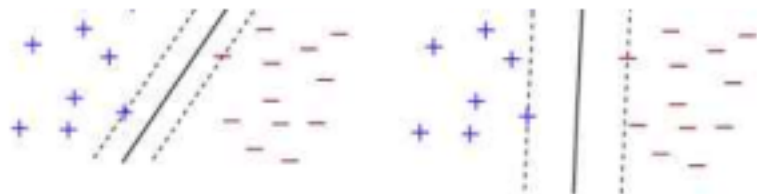
Which hyperplane w and b ?



Which hyperplane w and b ?



SVM



SVM optimization: maximizing the margin between + and -

Def.: Margin = distance between the hyperplanes $f(x) = 1$ and $f(x) = -1$ (dashed lines in Figure).

Intuitively, a classifier with a larger margin is more robust to fluctuations

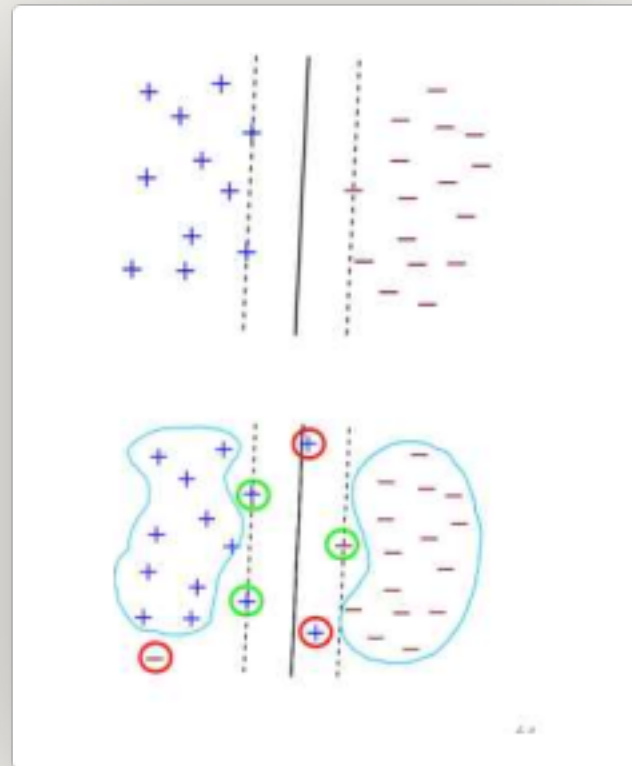
Hard Margin

Final expression for the Hard Margin SVM optimization:

$$\min_{w,b} P(w,b) = \frac{1}{2} \|w\|^2 \quad \text{with} \quad \forall i \quad y_i f(x_i) \geq 1$$

SVM

- **Hard margin:** OK if data are linearly separated
- Otherwise: noisy data (in red) disrupt the optimization
- **Solution: soft SVM**



SVM: soft margin

Introducing the slack variables ξ_i , one usually gets rid of the inconvenient max of the loss and rewrites the problem as

$$\min_{w,b} P(w,b) = \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad \text{with} \quad \begin{cases} \forall i & y_i f(x_i) \geq 1 - \xi_i \\ \forall i & \xi_i \geq 0 \end{cases}$$

Very large C → hard margin

Minimize $\|w\|$ (margin maximization) under the constraint that all training examples are correctly classified with zero loss.

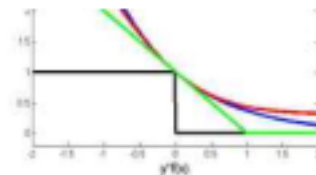
Smaller C → soft margin

Relaxes this constraint; SVMs produce markedly better results on noisy problems.

SVM learning scheme

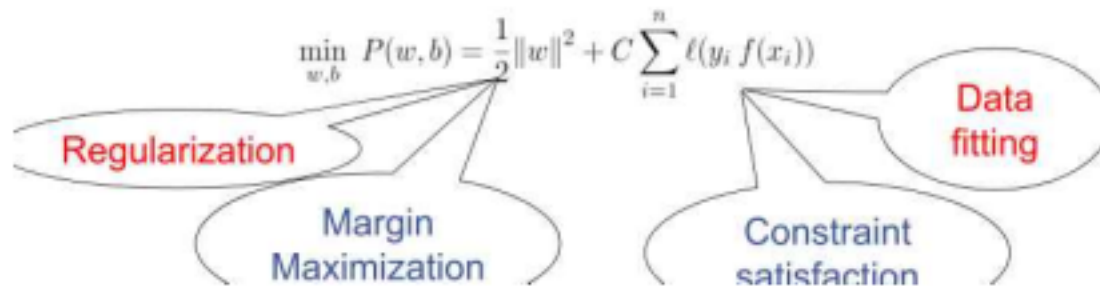
Regularization $\leftrightarrow$ Margin
maximization

Data fitting $\leftrightarrow$ Constraints



Equivalently, minimizing the following objective function in feature space with the hinge loss function:

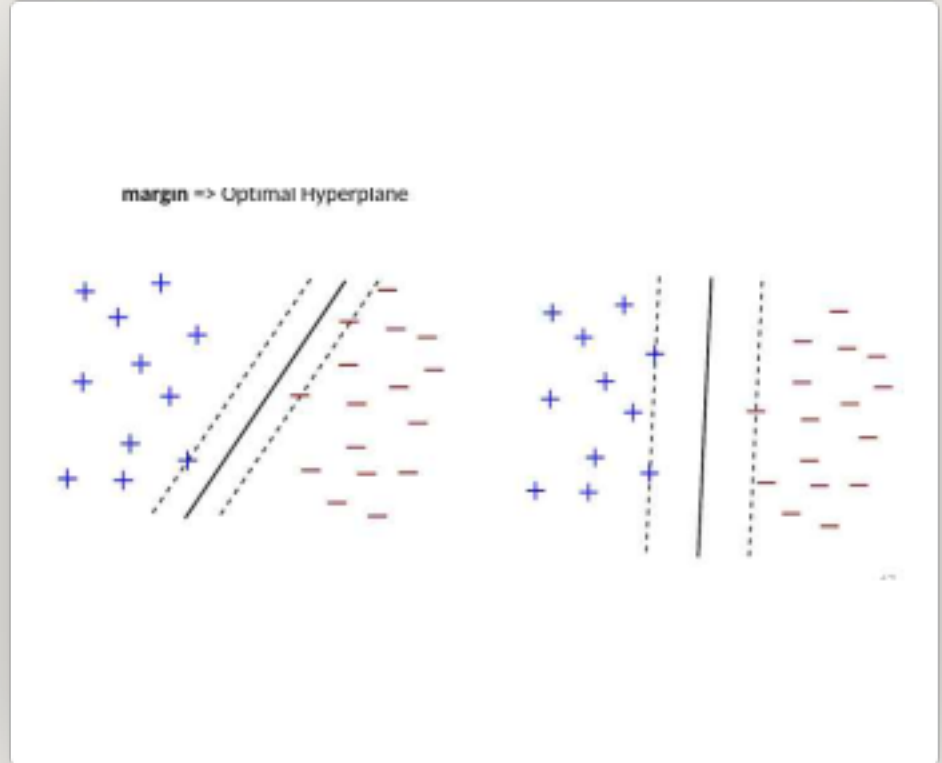
$$\ell(y_i f(x_i)) = \max(0, 1 - y_i f(x_i))$$



Solving equation: SVM

SVMs are defined by three incremental steps:

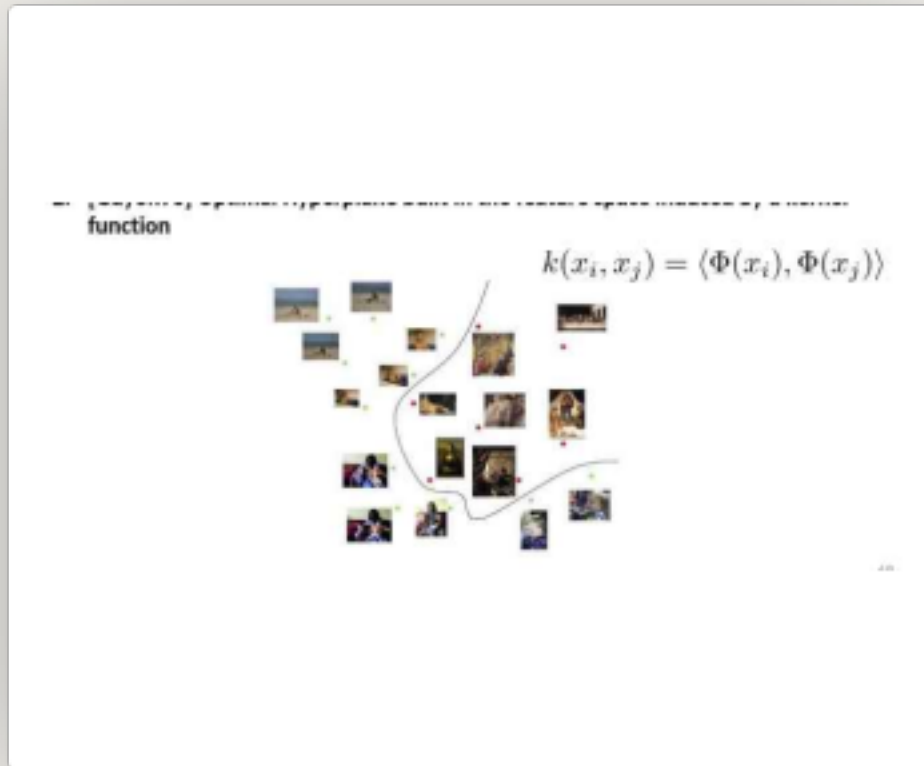
- 1 **[Vapnik63]** Linear classifier: separates the training examples with the widest margin → optimal hyperplane



Solving equation: SVM

SVMs are defined by three incremental steps:

- 1 **[Vapnik63]** Linear classifier: separates the training examples with the widest margin → optimal hyperplane
- 2 **[Guyon93]** Optimal hyperplane built in the feature space induced by a kernel function



Learning SVMs: primal / dual

- Primal optimization: $f(x) = \langle w, \Phi(x) \rangle + b$
- Dual optimization: learning SVMs can be achieved by solving the dual of this convex optimization problem
- Dual (using Lagrangian and $k(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle$):
 - For Hard Margin:

$$\max_{\alpha} \mathcal{L}(\alpha) = \sum_{i=1} \alpha_i - \frac{1}{2} \sum_{i,j} y_i y_j \alpha_i \alpha_j k(x_i, x_j) \quad \text{with} \quad \begin{cases} \sum_i \alpha_i y_i = 0 \\ 0 \leq \alpha_i \end{cases}$$

- For Soft Margin:

$$\max_{\alpha} \mathcal{L}(\alpha) = \sum_{i=1} \alpha_i - \frac{1}{2} \sum_{i,j} y_i y_j \alpha_i \alpha_j k(x_i, x_j) \quad \text{with} \quad \begin{cases} \sum_i \alpha_i y_i = 0 \\ 0 \leq \alpha_i \leq C \end{cases}$$

SVM optimization

Standard equivalent formulation without enforcing α_i to be positive:

- Optimization on coefficients α_i of the SVM kernel expansion $f(x) = \sum_{i=1}^n \alpha_i k(x, x_i) + b$ by defining the dual objective function:

$$D(\boldsymbol{\alpha}) = \sum_i \alpha_i y_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j k(x_i, x_j)$$

- and solving the SVM dual *Quadratic Programming* (QP) problem.

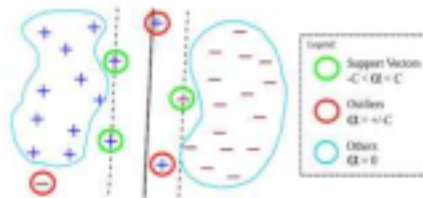
$$\max_{\boldsymbol{\alpha}} D(\boldsymbol{\alpha}) \quad \text{with} \quad \begin{cases} \sum_i \alpha_i = 0 \\ A_i \leq \alpha_i \leq B_i \\ A_i = \min(0, Cy_i) \\ B_i = \max(0, Cy_i) \end{cases}$$

Solving equation: SVM

SVMs are defined by three incremental steps:

- 1 **[Vapnik63]** Linear classifier: separates the training examples with the widest margin $\rightarrow$ optimal hyperplane
- 2 **[Guyon93]** Optimal hyperplane built in the feature space induced by a kernel function
- 3 **[Cortes95]** Soft version: noisy problems addressed by allowing some examples to violate the margin constraint

3. **[Cortes95]** soft version: noisy problems addressed by allowing some examples to violate the margin constraint



Solving the SVM

Min P or max D → quadratic programming (QP)

Good news: efficient batch numerical algorithms solve the specific SVM QP problem (hinge loss, convex objective, ...)

STRATEGIES EXPLOITING ITS SPECIFICS

- Conjugate gradient method [Vapnik]
- Sequential Minimal Optimization (SMO) [Platt]

Both use successive searches along well-chosen directions.

SOLVERS

- SVMLight [Joachims], SVMtorch: decomposition algorithms to define such directions
- SVMstruct, for structured outputs
- State-of-the-art SMO implementation: **libsvm** — used in tutorials
- LibLinear for primal optimization (with MATLAB)

SMO algorithm for SVM optimization

1. Set $\alpha \leftarrow \mathbf{0}$ and compute the initial gradient $\mathbf{g}$ of $D(\alpha)$
2. Choose a τ -violating pair(*) (i, j) Stop if no such pair exists
3. $\lambda \leftarrow \min \left\{ \frac{g_i - g_j}{k_{ii} + k_{jj} - 2k_{ij}}, B_i - \alpha_i, \alpha_j - A_j \right\}$
4. $\alpha_i \leftarrow \alpha_i + \lambda$, $\alpha_j \leftarrow \alpha_j - \lambda$
5. $g_s \leftarrow g_s - \lambda(k_{is} - k_{js}) \quad \forall s \in \{1 \dots n\}$
6. Return to step 2

(*) pairs in +1/-1 with significant diff of gradients

A ways to easily satisfy the null sum coeff constraint

A way to easily satisfy the null-sum coefficient constraint

SUMMARY

Classification pipeline

